

Source Exclusion in Regime-Gated Forecasting: A Cross-Market Audit of Equity-Factor Predictability

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A source can reach the forecast by two routes

The gate chooses how much each regime-specific regression contributes.

Explicit lag route

Past source-factor returns enter the regression directly through lag coefficients.

Regime-gate route

The source helps fit or update hidden-state probabilities. Those weights affect the forecast even without source-lag coefficients.

Whole pipeline

Preprocessing, scaling, validation, and tuning can also depend on the source. Complete exclusion must audit each route.

- A forward filter still uses future information if its parameters were fitted on future observations.

Three comparisons answer three different questions

Name the removal operation before interpreting its gain.

Comparison	What changes?	Question answered
Shared-gate ablation	Remove source lags; retain source-informed gate	Do explicit lags add value given this gate?
Complete source exclusion	Remove source from every restricted-model path	Does the full source-informed procedure improve forecasts?
Gate-value comparison	Gated versus pooled, same raw information	Does this fitted gate earn its extra complexity?

- Shared-gate ablation is valid for its own target; it is not complete information exclusion.

Negative fitted gain does not prove no information

The oracle value of additional information and the performance of fitted models are different objects.

Fitted gain = oracle information gain + restricted-model error – unrestricted-model error

- $\mathcal{R}$: expected squared loss; R and U denote source-excluded and source-included forecasts.
- E_R and E_U : excess risk relative to each oracle, including approximation and estimation error.
- Oracle information gain is nonnegative for nested information sets.
- Extra fitted-model error can outweigh information; a positive sample gain alone also does not establish population usefulness.

Evaluate the complete cross-market task family

Six factors, all 30 ordered directions, four regional panels.

Markets and outcomes

US, Europe, Japan, and Asia Pacific excluding Japan. One-day mean-return forecasts; two lagged observations in the frozen expansion.

Chronology

Train before 2005; validate 2005–2009; annually refit during evaluation. Report 2010–2024 and January 2025–August 2026 separately.

Inference family

4 regions $\times$ 30 directions $\times$ 5 procedure comparisons $\times$ 4 weight choices = 2,400 temporal endpoints.

- The later era was outside earlier repository forecast evaluation; globally untouched exposure is unverified.
- Panels share shocks and a provider; they are not independent replications.

Removal choice changes some point-estimate signs

Temporal capped score = $\min(\text{squared forecast error} / c, 1)$; c is fixed from training.

Quantity	Observed result	Interpretation
Unweighted HMM sign disagreements	42 / 240	Descriptive removal sensitivity
Directional temporal endpoints	0 / 2,400	No simultaneous directional crossing
Intervals within ± 0.005	776 / 2,400	All explained by forecast proximity
Exact matches among those 776	196	Zero paired loss

- Gain = restricted loss – source-included loss. Positive favors inclusion.
- In all 42 sign disagreements, both component gains lie within ± 0.005 . Median absolute direct contrast: 0.000341.

The original HML → SMB reconstruction is unfavorable

2010–2024, 3,774 matched dates; two-lag RMSE in basis points, lower is better.

Procedure	Exclude HML	Include HML
Pooled ridge	64.177	64.248
Gaussian HMM	64.319	64.403
Student-t HMM	64.257	64.322
Zero-return benchmark	63.948	63.948

- HML: value-minus-growth spread; SMB: small-minus-big spread.
- This current-vintage reconstruction is not an exact reproduction of the earlier study.
- A lag count does not establish an advance-warning horizon.

A controlled example separates pathways from fitting costs

Simulation: uncapped conditional squared-loss gains, in target-squared units; mean across 100 seeds.

Procedure	Complete-exclusion gain	Shared-gate gain
Exact oracle	0.1301	0.0000
Gaussian gate	0.1345	−0.0036
Student-t gate	0.1374	−0.0036

- The source carries information through a constructed state pathway.
- Keeping that pathway makes exact shared-gate gain zero.
- Small negative fitted shared-gate gains reflect procedure costs, not negative information.

An informative noisy source can still hurt fitted HMMs

Simulation: uncapped conditional squared-loss gains, in target-squared units; positive favors inclusion.

Source flip probability	Oracle gain	Gaussian-HMM gain	Student-t-HMM gain
0.0	0.1336	0.1386	0.1419
0.1	0.0708	-0.0494	-0.0486
0.3	0.0174	-0.0216	-0.0327
0.5	0.0000	-0.0336	-0.0446

- 100 independent seeds per noise level; seeds are paired across levels.
- Combined approximation, estimation, optimization, and tuning costs can outweigh information value.

Information paths and inference targets

The supported contribution is a reproducible procedure audit, not a validated warning system.

- Specify whether the source is removed from lags, the gate, or the entire pipeline.
- Keep information value separate from fitted-model performance.
- Report complete families, uncertainty, and discrepancy magnitudes.
- Capped-loss evidence does not establish ordinary-MSE coverage or trading value.

Source: github.com/ChorokLeeDev/regime-predictability-revision

Source exclusion in regime-gated forecasts